

Automated F1 Race Data Analysis & Power BI Dashboard Integration

Project Documentation | Updated October 2026

1. Project Overview

This project builds an end-to-end pipeline that takes raw Formula 1 timing and telemetry data and turns it into an interactive Power BI dashboard. The pipeline is fully automated from ingestion through to the analysis layer: pulling a new race into the dashboard requires re-running the notebook against a new event, not rebuilding any downstream logic. The report covers lap-time pace, speed, consistency and tyre strategy; Section 9 summarises what the data shows.

Pipeline at a glance: FastF1 API > Python analysis layer (pandas + custom OOP classes) > PostgreSQL > Power BI

2. Objectives

- Automate acquisition and processing of session data (laps, telemetry, positions, pit stops) via the FastF1 API
- Compute race-analysis metrics (sector pace, tyre strategy, pit stop performance, head-to-head comparisons) through a reusable, object-oriented analysis layer
- Persist processed results centrally in PostgreSQL rather than recomputing per session, with a self-updating view that incorporates new races automatically
- Surface the data through an interactive Power BI dashboard with overview, driver, team and tyre/stint views

3. Technical Stack

Layer	Tools
Data source	FastF1 API (session laps, telemetry, weather, timing)
Processing	Python 3.10, pandas, numpy
Persistence	PostgreSQL, SQLAlchemy, psycpg2
Visualization (exploratory)	matplotlib
Dashboard	Power BI Desktop (Power Query, DAX)
Environment	Dedicated conda environment (f1analysis) with a registered Jupyter kernel

4. Data Scope

The pipeline currently holds six races: **Abu Dhabi GP (2023, 2024, 2025)** and **Miami GP (2023, 2024, 2025)**, analyzed year-over-year at the same two circuits. This is a proof-of-pipeline stage, not a full-season dataset — the architecture is built to scale to a full season by re-running the same notebook against additional events, with no changes required to the database schema, the union view, or the Power BI report.

The report's table holds 6,602 laps (1,005 to 1,157 per race), with 20 drivers and 10 teams in every race. It contains 13 team names across the three seasons because of team rebrandings (for example AlphaTauri, RB and Racing Bulls).

5. Pipeline Architecture

5.1 Data Ingestion

FastF1 pulls session data directly from official F1 timing feeds. Sessions are cached locally so repeat runs against the same event do not re-hit the API.

5.2 Analysis Layer

A set of purpose-built Python classes wraps FastF1's raw session objects and produces analysis-ready tables and plots:

- **SectorStats** — per-driver sector time breakdown
- **SpeedStats** — speed-trap and top-speed analysis
- **TyreStrategyAnalysis / StintPerformance** — compound usage and stint length per driver
- **PitStopAnalysis** — pit stop count and duration
- **DriverLapData / LapComparison** — lap-by-lap and driver-vs-driver time deltas
- **TrackMap** — gear-shift and speed maps plotted along track XY coordinates
- **RacePosition** — lap-by-lap race position tracking

5.3 Persistence

Each race's lap table is written to PostgreSQL (telemetry columns are dropped, and timing columns are stored as text and converted to durations and seconds in Power Query). A union view (`f1_all_sessions`) automatically detects and incorporates newly added race tables that follow the project's naming convention, so adding a race does not require manually rewriting the view.

5.4 Visualization

Power BI connects directly to PostgreSQL and imports the union view as a single flat table. Year, Circuit and Race columns are derived from the race table name, which allows year-over-year comparison at each circuit. Metrics are defined as DAX measures (Section 8).

6. Environment Setup Notes

During setup, importing FastF1 raised a numpy/pandas binary-incompatibility error (`ValueError: numpy.dtype size changed`). Root cause: two separate Python installs on the machine (a standalone Python 3.9 and Anaconda3) with packages split across both, because the Anaconda3 site-packages directory was not writable and pip was silently falling back to user-level installs.

Resolution:

- Created a dedicated conda environment (`f1analysis`, Python 3.10)
- Installed `fastf1`, `pandas`, `numpy`, `SQLAlchemy`, `psycopg2-binary`, and `matplotlib` together in that single environment to avoid ABI conflicts
- Registered the environment as a Jupyter kernel and switched the notebook to it

Note for reproducing this environment: install all dependencies in one pass inside a clean environment rather than layering installs across multiple Python locations.

7. Power BI Dashboard Structure

The report has six pages. Home is a navigation hub; ToolTip is a hidden page used as a hover tooltip elsewhere in the report.

Home

Landing page with navigation buttons to the analysis pages.

Overview

Race-level KPI cards — Race Winner, Fastest Lap, Average Lap Time, Top Speed (speed trap), Total Laps, Total Drivers, Total Teams — alongside a Race Selection slicer and a line chart of average lap time by lap number.

Driver Analysis

Driver-level detail: a data table (Driver, Driver Number, Team, Highest Speed, Average Lap Time), two bar charts (Fastest Lap by Driver; Lap Time Consistency by Driver), a line chart of lap time by lap number split by driver, and Driver Number / Race slicers.

Tyre & Stint Analysis

Tyre strategy and pace for one selected race: a line chart of average clean-lap time by tyre age with one line per compound, a stacked column chart of laps driven by each driver on each compound, a column chart of pit stops by lap number, and a compound summary table (laps driven, clean laps, average clean lap, average stint length).

Team Analysis

Team-level detail: a data table (Team, Average Lap Time, Highest Speed, Race Winner, Race), a bar chart of average lap time by team, and a Race slicer.

ToolTip (hidden)

A custom hover-tooltip card showing Driver, Team, and Average Lap Time, surfaced on other visuals in the report.

8. Metric Definitions

Lap-time metrics use **clean laps** unless stated otherwise. A clean lap is a lap FastF1 flags as accurate (5,939 of the 6,602 laps). In this data the flag excludes every pit-in and pit-out lap, and all 96 safety-car-flagged laps at Miami 2024.

Metric	Definition	Notes
Average Lap Time	Mean lap time in seconds over clean laps	Raw all-lap averages are inflated by pit and safety-car laps (Finding 1)
Lap Time Consistency	Population standard deviation of clean-lap time (lower = steadier)	Measures spread, so it also picks up fuel-burn and tyre trends; drivers with few laps are unstable
Fastest Lap	Minimum lap time in the selected race	In all six races the fastest lap is also accurate and not deleted
Top Speed (speed trap)	Maximum speed-trap reading (SpeedST)	A single measurement point, not true peak speed; slipstreaming raises readings
Pit Stops	Number of laps with a recorded pit-in time	May include cars that retire into the pits
Average Stint Length	Mean laps per driver stint within a race	Depends on the Stint field being complete (see Section 11)
Avg Clean Lap (min 5 laps)	Average clean-lap time, hidden when fewer than 5 clean laps	The 5-lap threshold is a judgement call
Race Winner	Driver in P1 on the final lap number of the selected race	Blank when more than one race is selected

9. Key Findings

Findings are computed from the 6,602-lap table behind the report, using the definitions in Section 8. Six races at two circuits support observations about those races, not season-long or sport-wide trends. Where a cause is suggested but not tested, it is labelled as a hypothesis.

Race	Winner (team)	Fastest lap	Avg clean lap	Top speed (trap)
Abu Dhabi 2023	VER (Red Bull Racing)	VER 1:26.993	90.15 s	335 km/h
Abu Dhabi 2024	NOR (McLaren)	MAG 1:25.637	89.50 s	341 km/h
Abu Dhabi 2025	VER (Red Bull Racing)	LEC 1:26.725	89.58 s	345 km/h
Miami 2023	VER (Red Bull Racing)	VER 1:29.708	92.78 s	348 km/h
Miami 2024	NOR (McLaren)	PIA 1:30.634	93.01 s	355 km/h
Miami 2025	PIA (McLaren)	NOR 1:29.746	92.24 s	345 km/h

Finding 1 — Using all laps badly distorts consistency

Across the six races, the within-race standard deviation of lap time was 2.87 to 8.88 s using all laps, but only 0.94 to 1.46 s using clean laps. The largest change was Miami 2024 (8.88 s to 1.05 s), where 96 laps were flagged as safety car and none of them is a clean lap. The same filter moves Miami 2024's average lap from 96.00 s to 93.01 s. Driver consistency rankings changed substantially: the rank correlation between the all-laps and clean-laps orderings was 0.43 at Miami 2024 and 0.19 at Miami 2023, so a consistency ranking built on raw laps mostly reflects who pitted or was caught by a safety car. For this reason the report's lap-time metrics use clean laps.

Finding 2 — Race pace moved by under 1 s a year at both circuits

At Abu Dhabi the average clean lap went 90.15 s (2023), 89.50 s (2024) and 89.58 s (2025): 0.65 s faster in 2024, then within 0.08 s. At Miami it went 92.78 s, 93.01 s and 92.24 s, with 2025 being 0.77 s faster than 2024. Race-average lap time also reflects fuel load, tyre choice, traffic, weather and track conditions, so these figures are not a measure of car development.

Finding 3 — The fastest team on average was not always the winning team

Red Bull had the lowest average clean lap at both 2023 races (0.36 s clear of second place at Abu Dhabi, 0.66 s at Miami) and won both. In 2024 the leaders were nearly level: Ferrari (88.77 s) and McLaren (88.78 s) were 0.01 s apart at Abu Dhabi, and Ferrari, McLaren and Red Bull were within 0.03 s at Miami. In 2025 McLaren had the lowest average at both circuits (by 0.35 s at Abu Dhabi, 0.71 s at Miami); at Miami 2023 McLaren's average (93.42 s) was the slowest in the field. The winning team had the lowest average in only 3 of the 6 races, and Red Bull won Abu Dhabi 2025 with the third-fastest average, 0.63 s behind McLaren. *Hypothesis (not tested)*: strategy, track position and safety-car timing explain the gap between average pace and results.

Finding 4 — Abu Dhabi stopped less and ran longer stints each year

Pit-in laps at Abu Dhabi fell from 38 (1.90 per driver) in 2023 to 30 (1.50) in 2024 and 27 (1.35) in 2025, while average stint length rose from 20.3 to 21.6 to 24.6 laps. The share of laps on HARD fell from 73.5% to 66.7% to 61.7% as MEDIUM rose from 26.4% to 30.7% to 36.3%. This is consistent with a shift from two-stop towards one-stop strategies, though the data does not record strategy directly. At Miami, 2023 had 20 pit-ins (every driver made exactly one stop) and 2024 had 28, and 2024 was the only race where MEDIUM covered more laps (50.6%) than HARD (45.5%). Miami 2025 is excluded because of missing tyre data (Section 11).

Finding 5 — Safety cars and virtual safety cars bunch the pit stops

At Miami 2024 the safety car is flagged on laps 28 to 32 and the virtual safety car on laps 22 to 23. Seven drivers pitted on lap 28, the highest single-lap pit count in any of the six races (no other race has more than four), and three pitted on lap 22. At Miami 2025 a virtual safety car is flagged on laps 28 to 34, with 7 pit-ins across laps 28 and 29. The other four races show no safety-car laps and no lap with more than three stops.

Finding 6 — No consistent compound pace ranking

Excluding Miami 2025, the average clean lap on HARD was lower than on MEDIUM in 3 of 5 races, by 1.01 s (Abu Dhabi 2023: 89.89 vs 90.90), 0.51 s (Abu Dhabi 2024) and 0.19 s (Miami 2024); MEDIUM was lower in the other two, by 0.22 s (Abu Dhabi 2025) and 0.10 s (Miami 2023). These gaps are small next to the other things that differ between stints (fuel load, track evolution, traffic), so they do not show that either compound is faster. SOFT has too few clean laps to compare (at most 37 in any race).

Finding 7 — Lap times did not rise with tyre age

Comparing HARD laps at tyre age 21+ with laps at tyre age 1 to 10: Abu Dhabi differed by +0.27 s (2023), -0.07 s (2024) and +0.05 s (2025), while Miami laps on older tyres were faster by 0.64 s (2023) and 1.11 s (2024). For MEDIUM in 2024 the differences were -0.33 s at Abu Dhabi and -1.25 s at Miami. No race shows a clear wear penalty. This does not mean the tyres did not wear: cars lighten as fuel burns off and the track evolves, both of which reduce lap times and can hide wear. No fuel correction was applied.

Finding 8 — Speed-trap readings

Maximum speed-trap readings rose each year at Abu Dhabi (335, 341, 345 km/h). At Miami they peaked at 355 km/h in 2024 and were 345 km/h in 2025. Each figure is a single measurement point and is affected by slipstreaming, so small differences between races or teams should not be over-read.

10. Hypotheses and Next Analyses

The data supports the observations above but not recommendations to any stakeholder. These are the analyses that would test the open questions:

- **Safety-car effect on results:** add TrackStatus to the report and test whether safety-car timing explains the gap between average pace and finishing position (Finding 3).
- **Fuel-corrected tyre degradation:** apply a per-lap fuel-effect adjustment so tyre wear can be separated from fuel burn (Finding 7).
- **Strategy classification:** count stops per driver from the stint data to confirm the shift from two-stop to one-stop at Abu Dhabi (Finding 4), and extend to more circuits to see if it generalises.
- **Recover the missing Miami 2025 tyre data** so that race can be included in the tyre findings.

11. Limitations & Next Steps

- **Sample size:** six races loaded so far, covering two circuits (Abu Dhabi, Miami) across 2023–2025 — findings should be read as year-over-year comparison at those circuits, not a full-season trend. Next step is expanding to additional circuits and rounds.
- **Analysis-layer coverage:** tyre, stint and pit-stop metrics are now in the report (computed in DAX from the lap table), but sector-time analysis and the telemetry views (TrackMap, gear and speed maps, RacePosition) are not yet represented.
- **Missing tyre data, Miami 2025:** compound and stint are blank for 354 laps (laps 1 to 24 for 15 of the 20 drivers). Those laps are excluded from the tyre findings, and the Average Stint Length and compound visuals for that race are incomplete and should not be read.

- **Driver_Analysis line chart:** the lap-by-lap line chart currently plots raw lap times (pit and safety-car laps included); switching it to the clean-lap measure is a pending change.
- **Small compound samples:** SOFT has at most 37 clean laps in any race, so no SOFT conclusions are drawn.
- **Database configuration:** the database name is set separately in the notebook classes, the union-view function and the Power BI connection; these are to be unified in a single configuration.
- **Credential management:** migrating database credentials from inline notebook variables to environment-variable / .env-based configuration, consistent with the pattern used in the AI PostgreSQL Analytics Assistant project.
- **Documentation:** adding inline markdown documentation to the notebook itself alongside this external write-up.

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